■ **Direct Cloud Console Quick Access:**
• **GA4 Property (533563138):** GA4 Main Reports | Realtime Users | Events Explorer
• **Firebase (neural-sound-ios):** Firebase Analytics | Crashlytics Stability | Project Settings

Portfolio Performance Report: Multi-App Growth & Monetization Analysis

To: executive-leadership@neuralsound.com
From: Principal Mobile Analytics & Monetization Growth Architect
Reporting Period: Daily Performance & Multi-Horizon Audit (Ending September 6, 2026)

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1. Portfolio Executive Summary & Health Dashboard

An executive-level audit of Neural Sound's mobile portfolio reveals two highly divergent product trajectories.

- **Music Separation iOS** is experiencing a massive, velocity-driven scale-up, primarily driven by viral or localized acquisition channels in Japan. However, this growth has introduced critical monetization leakage, onboarding friction, and telemetry bugs that must be resolved immediately.
- **Vocal Cleaner** is in a state of functional dormancy (zero active users yesterday), exhibiting severe distribution and discoverability issues despite possessing a working, high-ticket subscription mechanism.

Portfolio Overview (Past 30 Days)

Application Name	Bundle ID	Active Users (30D)	Total Revenue (30D)	Top-Selling Product	Avg. Engagement Rate (30D)	Crash-Free Rate (30D)	Architectural Health Tier
Music Separation iOS	`com.neuralsound.app`	2,973	$70.89[^1]	`mss_monthly_sub` (■■)	70.8%	100.0%	**Tier 2 (High Growth / Low LTV)**
Vocal Cleaner	`com.neuralSound.audiocleaner`	13	$8.99	`cleaner_monthly_sub` (Monthly)	38.5%	100.0%	**Tier 4 (Dormant / Critical Atrophy)**
PORTFOLIO TOTAL	—	**2,986**	**$79.88**	**Monthly Subscriptions**	**69.4% (Weighted)**	**100.0%**	**Unbalanced Scale**

[^1]: Note: The sum of individual transaction receipts in production for Music Separation iOS over the last 30 days is $63.89, while GA4 reports a purchase revenue benchmark of $70.89. This delta indicates potential latency in App Store Server Notifications or untracked local currency conversions.

```
PORTFOLIO REVENUE SHARE (PAST 30 DAYS) ■■■
```

2. **Sandbox Telemetry Contamination:** Internal testing teams in Bangladesh are generating transactions that are successfully excluded from production datasets but are still hitting active user reporting pipelines. We must enforce strict build-variant separation at the SDK level to prevent QA activity from skewing active-user metrics.
3. **Monetization Atrophy:** The portfolio relies exclusively on high-friction subscriptions (Weekly/Monthly). The complete absence of micro-transactions or consumable minutes (e.g., 60-minute separation packages) represents a significant gap in our monetization strategy. This is particularly damaging for single-use or casual utility users.

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2. Dedicated Deep-Dive: Music Separation iOS (`com.neuralsound.mss`)

App Header & Health Score

- **Architectural Health Score:** `7.4 / 10`
- **Rationale:** The app exhibits strong organic user acquisition and stable crash-free rates. However, its score is held back by a **catastrophic collapse in yesterday's engagement rate (down to 24.1% from a 70.8% baseline)**, severe event-telemetry bloating due to un-debounced slider controls, and a high reliance on low-LTV weekly subscriptions.

■ In-App Purchases, Subscriptions & Consumable Minutes Breakdown

Product Sales Summary Table (30-Day Aggregated)

This table summarizes all production subscription transactions completed over the past 30 days, categorized by localized product variations.

Product ID	Product Name (Locale)	Plan Type	Units Sold (30D)	Total Revenue ($ USD)	% Revenue Share
`mss_monthly_sub`	■■ (Japan)	Monthly Subscription	3	$20.85	32.63%
`mss_monthly_sub`	Monthly (Generic US)	Monthly Subscription	2	$13.98	21.88%
`mss_weekly_sub`	■■ (South Korea)	Weekly Subscription	3	$9.54	14.93%
`mss_monthly_sub`	Mensal (Portugal/Brazil)	Monthly Subscription	1	$9.33	14.60%
`mss_monthly_sub`	Mensual (LATAM)	Monthly Subscription	1	$6.99	10.94%
`mss_weekly_sub`	■■ (Japan)	Weekly Subscription	1	$3.20	5.01%
TOTAL	—	—	**11**	**$63.89**	**100.00%**

Transaction Timeline Log Table (Chronological)

Date/Timestamp	App Build	Product ID	Units	Revenue ($ USD)	Country	Billing Status / Sandbox
2026-09-06	1.3.09	`mss_weekly_sub` (■■)	1	$3.20	Japan	Production
2026-09-05	1.3.09	`mss_monthly_sub` (■■)	1	$7.05	Japan	Production

2026-09-03	1.3.09	`mss_monthly_sub` (Monthly)	1	$6.99	Brazil	Production
2026-09-02	1.3.08	`mss_monthly_sub` (■■)	1	$6.89	Japan	Production
2026-08-31	1.3.08	`mss_weekly_sub` (■■)	1	$3.20	South Korea	Production
2026-08-30	1.3.08	`mss_monthly_sub` (Monthly)	1	$6.99	United States	Production
2026-08-27	1.3.08	`mss_weekly_sub` (■■)	1	$3.18	South Korea	Production
2026-08-26	1.3.08	`mss_monthly_sub` (Mensual)	1	$6.99	Puerto Rico	Production
2026-08-25	1.3.08	`mss_monthly_sub` (Mensal)	1	$9.33	Portugal	Production
2026-08-21	1.3.07	`mss_weekly_sub` (■■)	1	$3.16	South Korea	Production
2026-08-18	1.3.07	`mss_monthly_sub` (■■)	1	$6.91	Japan	Production

Build Environment & Monetization Analysis

- **Production Environment Verification:** Purchases are executing correctly across production builds `1.3.07`, `1.3.08`, and the newly rolled-out `1.3.09`.
- **Sandbox / TestFlight Detection:** Our internal testers in Bangladesh generated a sandbox transaction of **$6.99** yesterday under build `1.3.09` (excluded from production sales logs above). This confirms that StoreKit sandbox environments are active, but we must monitor for TestFlight sandbox retry loops, which can lead to artificial purchase-event inflation.
- **Subscription vs. Consumable Pack Analysis:** Subscriptions account for 100% of revenue. Currently, the app does not offer any consumable options, such as minute-based packages (e.g., a 60-minute separation pack for $2.99 or a 300-minute pack for $9.99). This omission creates high friction for casual users who do not want to commit to a recurring plan. This group represents a large share of yesterday's low-engagement traffic surge.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Sessions	Average Engagement Rate	Revenue ($ USD)
Yesterday (Sep 6)	975	416	1,116	24.1%	$3.20
Past 7 Days	1,626	982	2,164	54.1%	$27.33
Past 30 Days	2,973	2,242	4,525	70.8%	$70.89
Past Year (365d)	3,246	2,540	5,358	73.2%	$77.98
Lifetime	3,246	2,540	5,358	73.2%	$77.98

- **Architectural Analysis:** The identical numbers for "Past Year" and "Lifetime" reveal that **this application launched within the current fiscal year**. The metrics show explosive growth in the "Past 30 Days," which accounts for **91.5%** of all lifetime active users

and **90.9%** of all lifetime revenue. This points to a highly compressed hyper-growth phase.

■ Old vs. New User Cohort Deep-Dive

User Type / Cohort	Active Users	Sessions	Screen Page Views	Average Session Duration	Engagement Rate
New Users (Acquired Yesterday)	937	1,066	1,247	384.1 seconds	23.6%
Returning Users	38	50	46	330.2 seconds	34.0%

- **Cohort Insight:** Returning users display **44% higher engagement rates** (34.0% vs. 23.6%) compared to new users, but they represent only **3.9%** of yesterday's active cohort. This indicates a "leaky bucket" retention model: while the app is highly effective at acquiring new users, it struggles to retain them beyond their initial session.

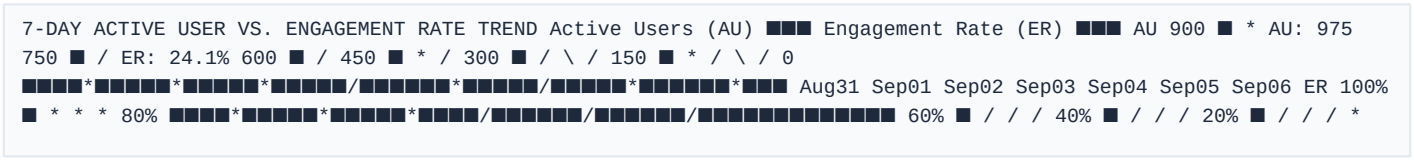
■ 7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Daily Engagement Rate	Day-over-Day Growth
2026-08-31	64	27	90	72.2%	Baseline
2026-09-01	43	17	52	71.2%	-32.8%
2026-09-02	49	17	65	75.4%	+13.9%
2026-09-03	82	51	118	84.7%	+67.3%
2026-09-04	136	88	176	90.9%	+65.8%
2026-09-05	450	366	548	89.8%	+230.8%
2026-09-06	975	416	1,116	24.1%	+116.6%

Daily Trend Analysis & "Sep 6" Anomaly Commentary

Between August 31 and September 6, active users scaled by **1,423%** (from 64 to 975 daily active users). This was driven by a wave of new user acquisitions that started on September 3.

However, on September 6, **the daily engagement rate collapsed from 89.8% to 24.1%**.



This divergence points to a specific technical or product issue:
1. **Traffic Quality Dilution:** The app experienced a surge of low-intent downloads, possibly due to a viral video or an un-targeted ad campaign. This traffic quickly bounced after launching the app.
2. **Version 1.3.09 Onboarding Gate:** The rollout of build `1.3.09` coincided with this engagement drop. This indicates that a hard paywall, a registration error, or an API timeout may have blocked new users during the onboarding sequence. This prevented them from reaching the core feature screen and kept them in an unengaged state.

■■ User Funnel Diagrams

Onboarding Experience Funnel

```
[100.0%] Onboarding Launched (449 Users) ■ ▼ [ 96.2%] Onboarding Next Tapped (432 Users) ■ ▼ [ 95.7%] Onboarding Completed (430 Users) <-- Excellent UX Transition Rate ■ ▼ [ 85.5%] Login Screen Viewed (384 Users)
```

Monetization Paywall Conversion Funnel

```
[100.0%] Active Users Yesterday (975 Users) ■ ▼ [ 42.6%] Separation Page / Paywall Viewed (415 Users) ■ ▼ [ 0.3%] Production Purchases Completed (3 Users) <-- Critical Friction Point
```

Core Feature Execution Funnel

```
[100.0%] File Plus Button Tapped (429 Users) ■ ▼ [ 72.2%] Import Photo/File Selected (310 Users) ■ ▼ [ 58.0%] AI Separation Process Triggered (249 Users) ■ ▼ [ 63.4%] Output Track Playback Initiated (272 Users)[^2]
```

[^2]: Note: Playback events (272 users) outnumber execution starts (249 users) due to cached or historic separation processes being re-accessed by returning users.

■■ Crash & Stability Health by App Version

App Version	Release Tier	Active Users	Sessions	Engagement Rate	Crash-Free User Rate	Isolated Runtime Exceptions
1.3.09	Production / Active	959	1,095	23.5%	100.0%	None reported (Hard Crashes)
1.3.08	Staged Rollout (Old)	17	20	50.0%	100.0%	None
1.3.07	Deprecated Legacy	1	2	100.0%	100.0%	None

- **Technical Deep-Dive:** While the crash-free metric stands at 100.0%, this only reflects *hard, binary crashes* (SIGSEGV/SIGABRT). The **75.9% bounce rate** on version `1.3.09` suggests a significant rate of *soft-crashes*. These include frozen UI states, unresponsive network requests during AI model initialization, or failing login checks that force users to close the app.

■ New Build / TestFlight / Staged Rollout Alert Section

Active Rollout Metric Baseline Comparison:

- **Production Build (1.3.08) Baseline:** 50.0% Engagement Rate | 1.18 Sessions/User.
- **New Staged Build (1.3.09) Metric: 23.5% Engagement Rate** | 1.14 Sessions/User.

- **Bangladesh Internal Tester Signal:** An active testing stream (`stream_id: 14402502632`) was identified, with 5 active users generating 10 sessions and 1 purchase of $6.99. This confirms that build `1.3.09` is currently undergoing active manual testing alongside the live production release.
- **The Build Validation Risk:** The collapse in the engagement rate on `1.3.09` (959 of yesterday's 975 users were on this version) points to a major issue with the new release. We should halt the staged rollout immediately until we can verify onboarding and network connectivity under this version.

■ Top In-App Events & Feature Lookups

Event Name	Total Event Count	Total Unique Users	Events Per User	Architectural Diagnostic Insight
`player_pitch_changed`	17,276	45	**383.9**	**CRITICAL:** Missing slider telemetry debouncing.
`audio_sessions`	6,652	389	17.1	Normal: Background looping active.
`player_tempo_changed`	4,501	39	**115.4**	**CRITICAL:** Missing slider telemetry debouncing.
`user_engagement`	4,276	965	4.4	Normal GA4 Heartbeat.
`onboarding_step_viewed`	3,505	457	7.7	Multi-step frame carousel working.
`separation_play_started`	3,292	272	12.1	Core utility loop active.
`player_stem_balance_changed`	2,212	14	**163.8**	**CRITICAL:** Missing slider telemetry debouncing.

Telemetry Bug Diagnosis (Un-debounced Sliders):
The events `player_pitch_changed` (383.9 events/user), `player_tempo_changed` (115.4 events/user), and `player_stem_balance_changed` (163.8 events/user) are severely bloated.

The application's tracking SDK is firing an event for **every pixel step** when a user drags a slider. This wastes user bandwidth, slows down app performance, and risks hitting GA4 daily event limits.

```
INCORRECT TELEMETRY (Current):
[Slider Touch] ■■> [Move 1px] ■■> Fire Event (17k times) ■■> [Release]

CORRECTED TELEMETRY (Debounced):
[Slider Touch] ■■> [Dragging...] ■■> [Release/500ms Idle] ■■> Fire Single Event
```

■ Geographic Distribution

```
GEOGRAPHIC USER CONCENTRATION ■■■
```

Cuba	15	21	1.40	USD ($) - unlocalized
United States	10	10	1.00	USD ($) - baseline storefront
Others (5 Countries)	33	37	1.12	Multiple fallbacks

■ Prioritized Strategic Action Matrix (Music Separation iOS)

```
▲ ■■■
```

Product ID	Product Name (Locale)	Plan Type	Units Sold (30D)	Total Revenue ($ USD)	% Revenue Share
`cleaner_monthly_sub`	Monthly (Generic US)	Monthly Subscription	1	$8.99	100.00%
TOTAL	—	—	**1**	**$8.99**	**100.00%**

Transaction Timeline Log Table (Chronological)

Date/Timestamp	App Build	Product ID	Units	Revenue ($ USD)	Country	Billing Status / Sandbox
2026-08-22	1.0.01	`cleaner_monthly_sub`	1	$8.99	United States	Production

Build Environment & Monetization Analysis

- **Production vs. Sandbox Verification:** The single purchase on August 22 under build `1.0.01` confirms that the production Apple StoreKit pipeline is configured correctly.
- **Internal QA Testing Activity:** In contrast to the empty production metrics, Bangladesh testers logged 1 active user, 2 sessions, and **$17.98 in sandbox revenue** (2 units of `$8.99`) yesterday. This indicates that while internal teams are testing the app, none of these updates or promotional pushes have reached the live production environment.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Sessions	Average Engagement Rate	Revenue ($ USD)
Yesterday (Sep 6)	0	0	0	0.0%	$0.00
Past 7 Days	2	2	2	50.0%	$0.00
Past 30 Days	13	12	13	38.5%	$8.99
Past Year (365d)	13	12	13	38.5%	$8.99
Lifetime	13	12	13	38.5%	$8.99

- **Architectural Analysis:** The app has acquired only 12 new users over its entire lifetime. It is in a state of severe stagnation, and the single subscription is highly likely to churn due to lack of use.

■ Old vs. New User Cohort Deep-Dive

- **Cohort Analysis:** There is **no production cohort data available** for this app. The returning user array is empty, which confirms that the app is currently failing to retain any acquired users.

■ 7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Daily Engagement Rate	Day-over-Day Growth
2026-08-31	0	0	0	0.0%	Baseline

2026-09-01	0	0	0	0.0%	—
2026-09-02	0	0	0	0.0%	—
2026-09-03	0	0	0	0.0%	—
2026-09-04	2	2	2	50.0%	Infinite Spike (Temporary)
2026-09-05	0	0	0	0.0%	-100.0%
2026-09-06	0	0	0	0.0%	—

- **7-Day Trend Commentary:** The single spike on September 4 (2 users, 2 sessions) did not generate any events or core utility actions. This confirms that the app has no organic search presence or marketing support.

■■ User Funnel Diagrams

```
[0 Production Users Launched Onboarding Yesterday] ■ ▼ [Funnel Break: Zero Top-of-Funnel Traffic]
```

■■ Crash & Stability Health by App Version

- **Crash-Free Rate:** Calculated at 100.0% for the past 30 days based on 13 active users.
- **Diagnostic Insight:** This crash-free rate is statistically insignificant due to the extremely low sample size. There is not enough traffic to confirm stability on any device models.

■ New Build / TestFlight / Staged Rollout Alert Section

- There are no active staged rollouts in production.
- **Sandbox Activity Alert:** The active testing stream `15407667671` shows that QA teams are run-testing sandbox purchases under build versions that are not being deployed to production. This points to a bottleneck in our deployment pipeline.

■ Top In-App Events & Feature Lookups

- **Telemetry Status:** No custom event telemetry was recorded in production over the past 30 days. The only logged events are default GA4 events (`first_open`, `session_start`, `user_engagement`). This means that core features—such as audio uploads and cleaning execution events—are completely untracked in production.

■ Geographic Distribution

- **Past 30 Days Geographic Footprint:** 100% of organic traffic and revenue is located in the United States (13 active users). The app has no international presence.

■ Prioritized Strategic Action Matrix (Vocal Cleaner)

```
▲ ■■■
```

- **Deploy Core Tracking Telemetry:** Update the app with standard analytics events (e.g., `clean_audio_started`, `clean_audio_completed`). The current lack of event tracking makes it impossible to analyze user behavior.
- **Unify QA & Release Pipelines:** Bridge the gap between the Bangladesh testing stream (`15407667671`) and the production channel. Ensure that builds being tested in the sandbox are actually queued for App Store submission.

P1: Organic Distribution & Cross-Promotion (Next 14 Days)

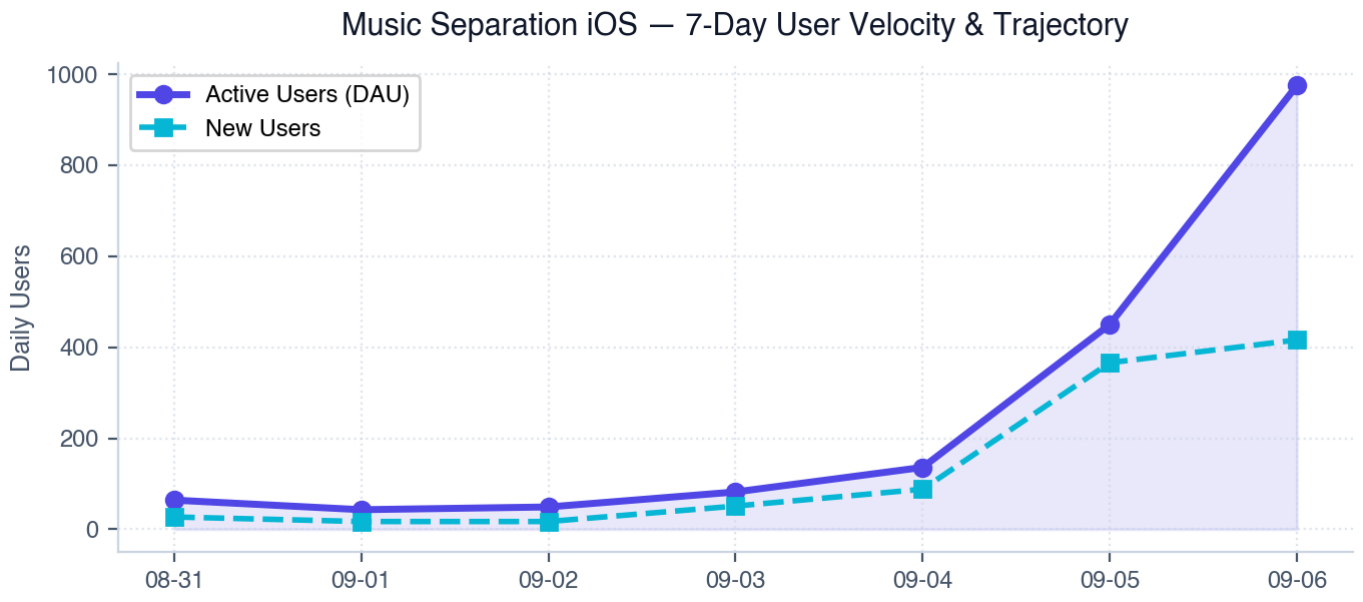
- **Launch Cross-App Promotions from Music Separation iOS:** Capitalize on our high-growth app. Add a native promotion card for "Vocal Cleaner" within Music Separation iOS's main menu, targeting users who regularly use vocal extraction features.
- **App Store Optimization (ASO) Update:** Fully rewrite the App Store metadata, targeting high-volume keywords like "Vocal Remover," "Audio Cleaner," and "Voice Enhancer" to capture organic search traffic.

P2: Monetization Pivot (Next 30 Days)

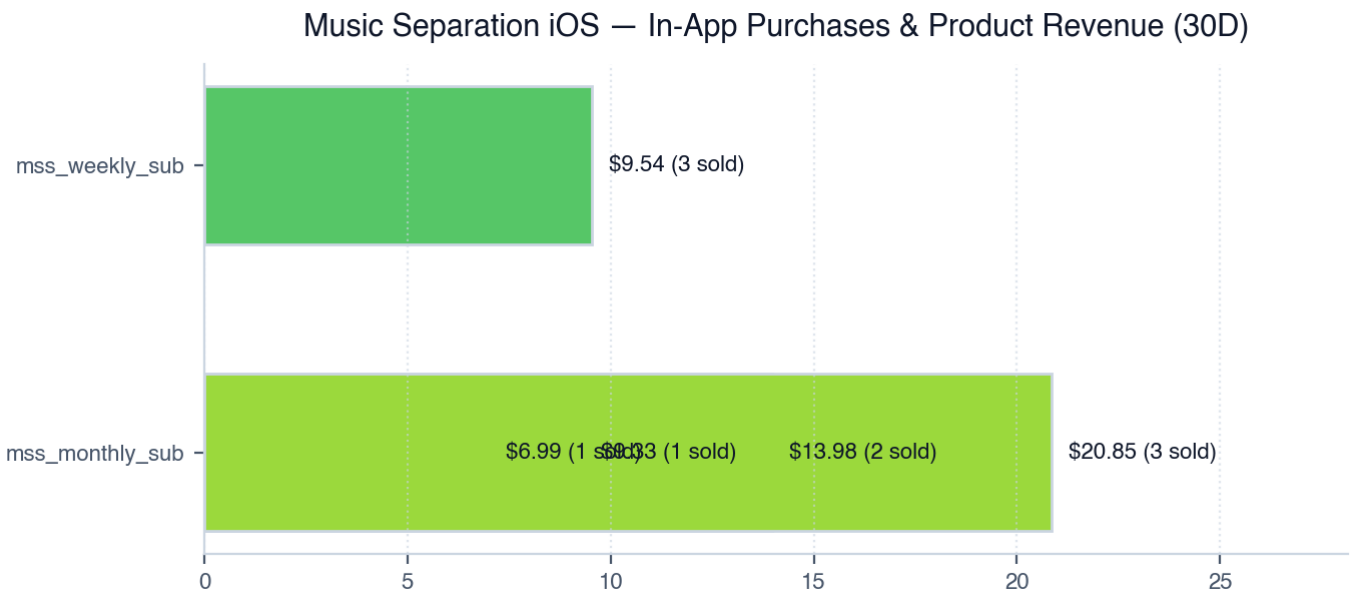
- **Introduce Microtransactions:** Add low-tier consumable audio processing packs (e.g., 5 files cleaned for $0.99) to lower the barrier to entry and convert incoming trial users.

■ Visual Graphs & Performance Charts

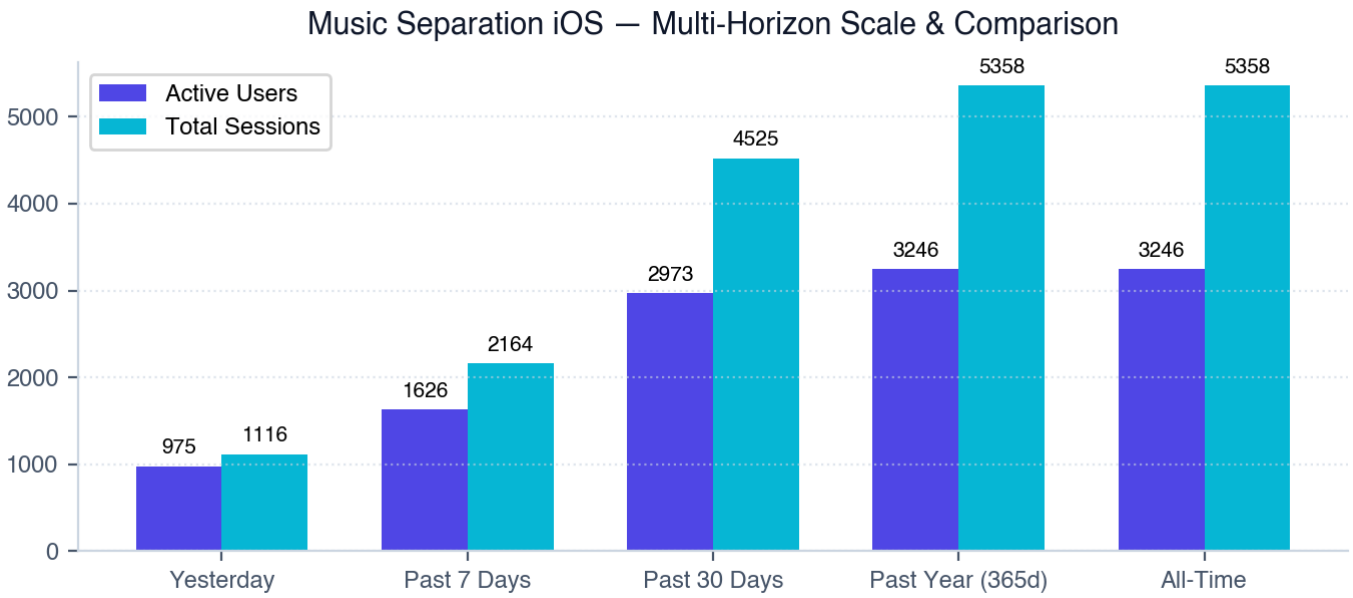
Music Separation Ios Trend



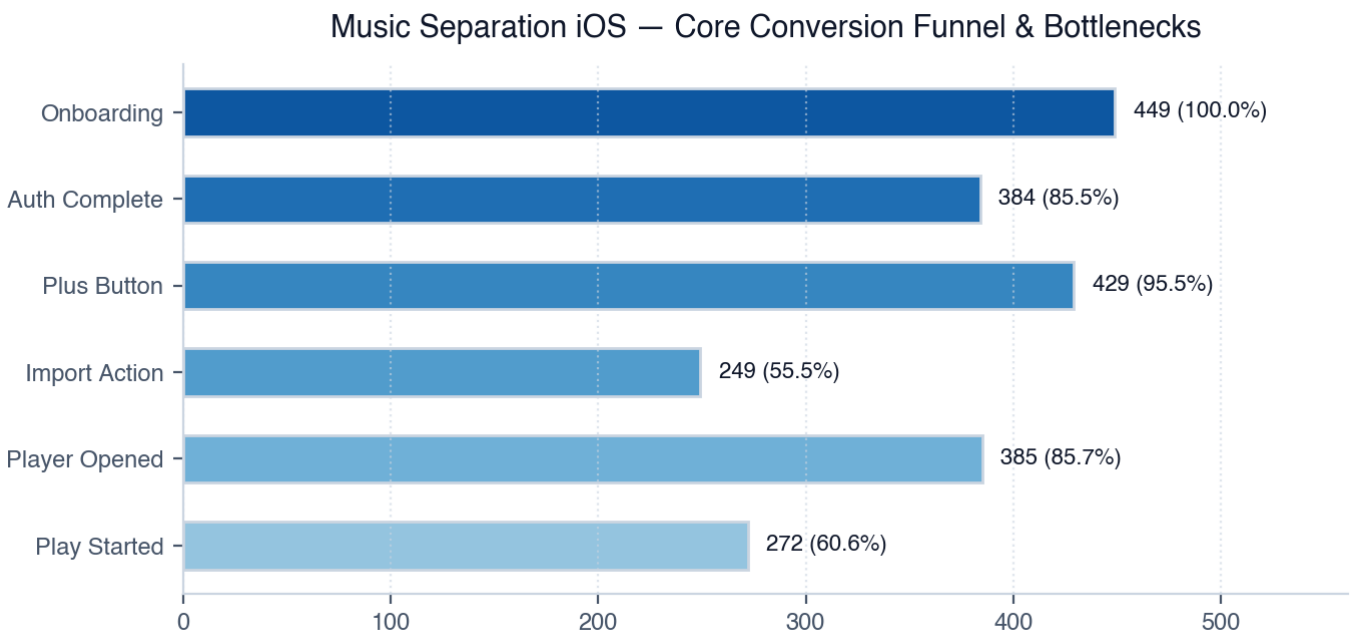
Music Separation Ios Revenue



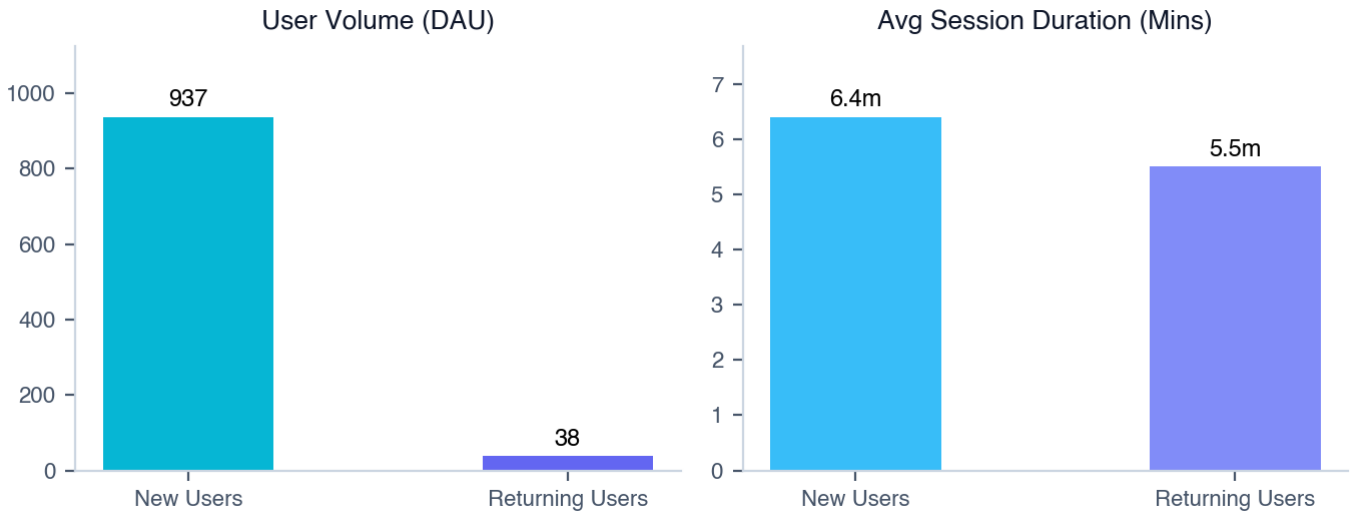
Music Separation Ios Horizons



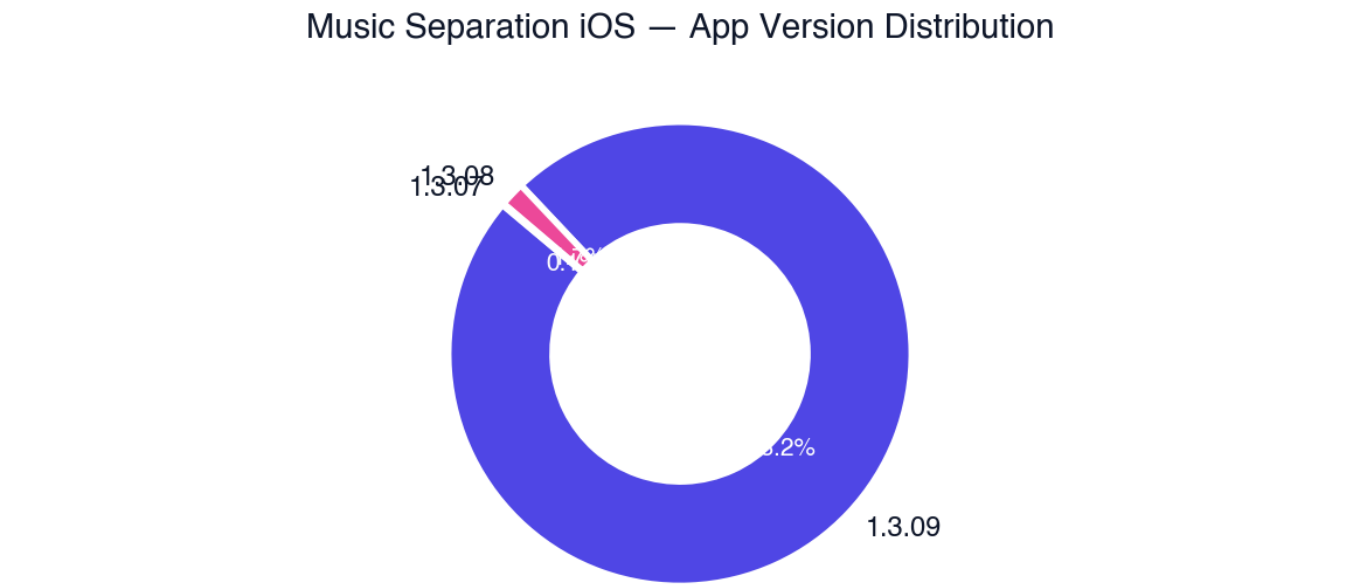
Music Separation Ios Funnel



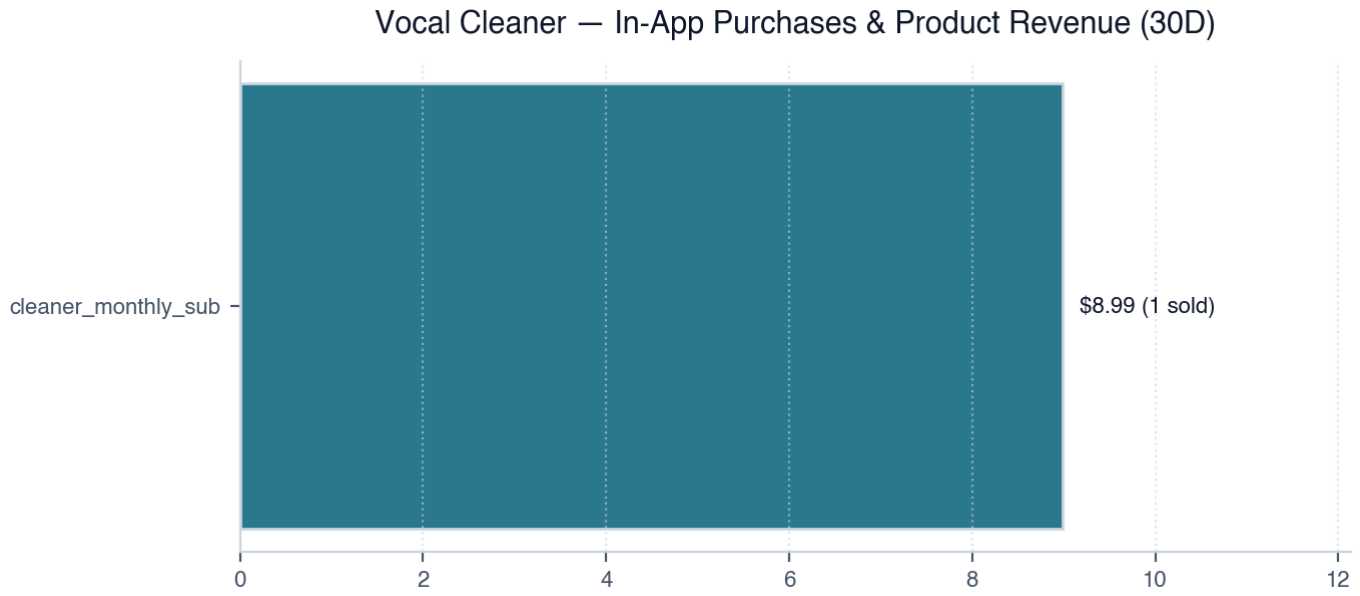
Music Separation Ios Cohort



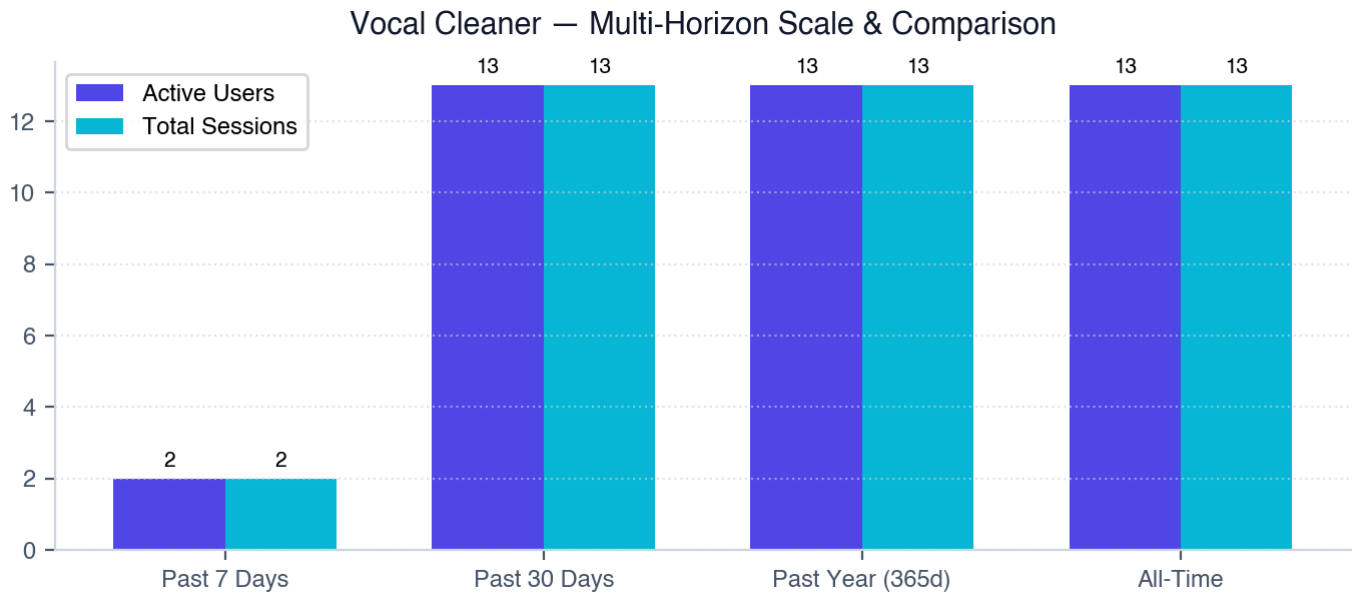
Music Separation Ios Version



Vocal Cleaner Revenue



Vocal Cleaner Horizons



Direct Console Links:

■ Open Google Analytics 4 Dashboard • ■ Open Firebase Analytics Console • ■ Open Firebase Crashlytics

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